Experiment 8

Report Scheduling, Automation and Alerts using Microsoft Power BI

Aim: To design and implement report scheduling, automation, and alert mechanisms using Microsoft Power BI and develop strategies for automated report delivery, monitoring, and business performance tracking.

OBJECTIVES:

1. To understand report scheduling and automation features in Microsoft Power BI.
2. To design scheduled reporting plans for daily, weekly, and monthly reports.
3. To develop automated report distribution strategies.
4. To configure KPI alerts for monitoring business performance.
5. To analyze monitoring scenarios and recommend corrective actions.

TOOLS / SOFTWARE REQUIRED:

1. **Operating System:** Windows 11.
2. **Software:** Microsoft Power BI Desktop.
3. **Dataset**: Global Superstore dataset in CSV or Excel format.
4. Power Query Editor
5. DAX (Data Analysis Expressions)
6. **System Requirements:** Minimum 4 GB RAM.

DATASET USED: Global Superstore Dataset.xlsx.

THEORY:

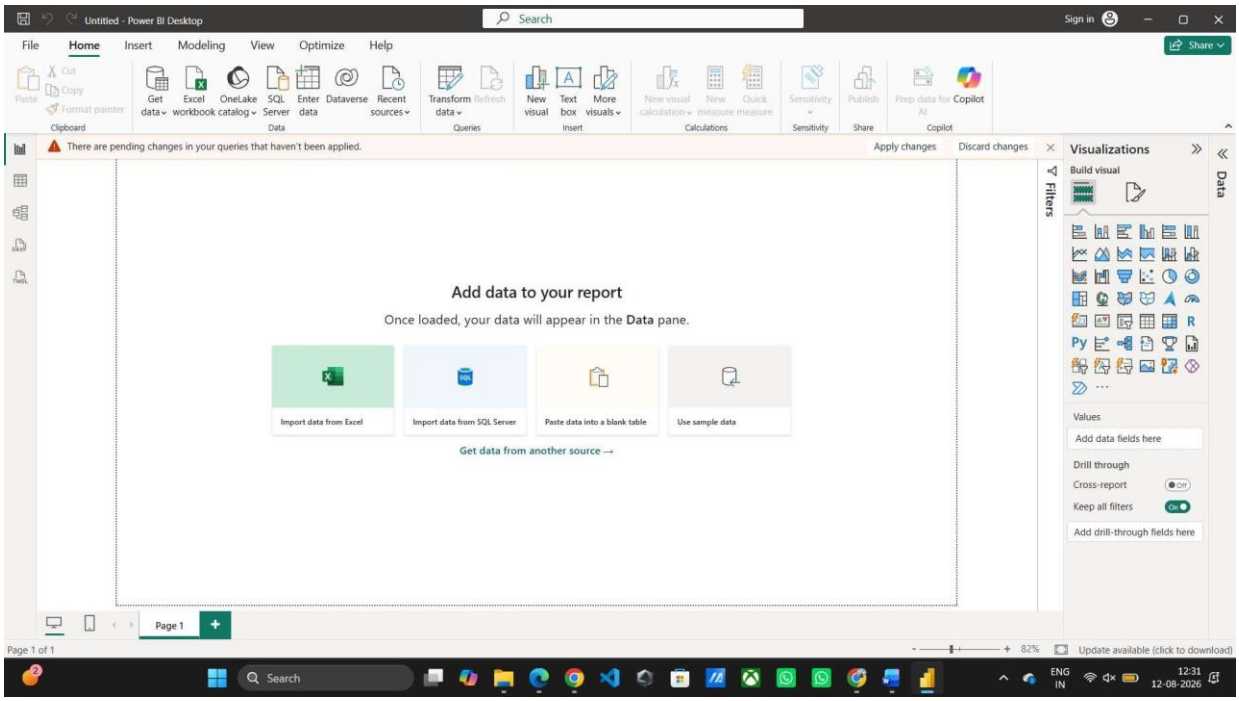
Business Intelligence systems must deliver information to stakeholders at the right time. Scheduled reporting, automated distribution, and KPI alerts help organizations monitor performance and take timely actions.

Power BI Service provides features such as subscriptions, scheduled refreshes, report sharing, and data-driven alerts. These features help organizations automate report delivery, monitor key performance indicators, and support effective business decision-making.

PROCEDURE:

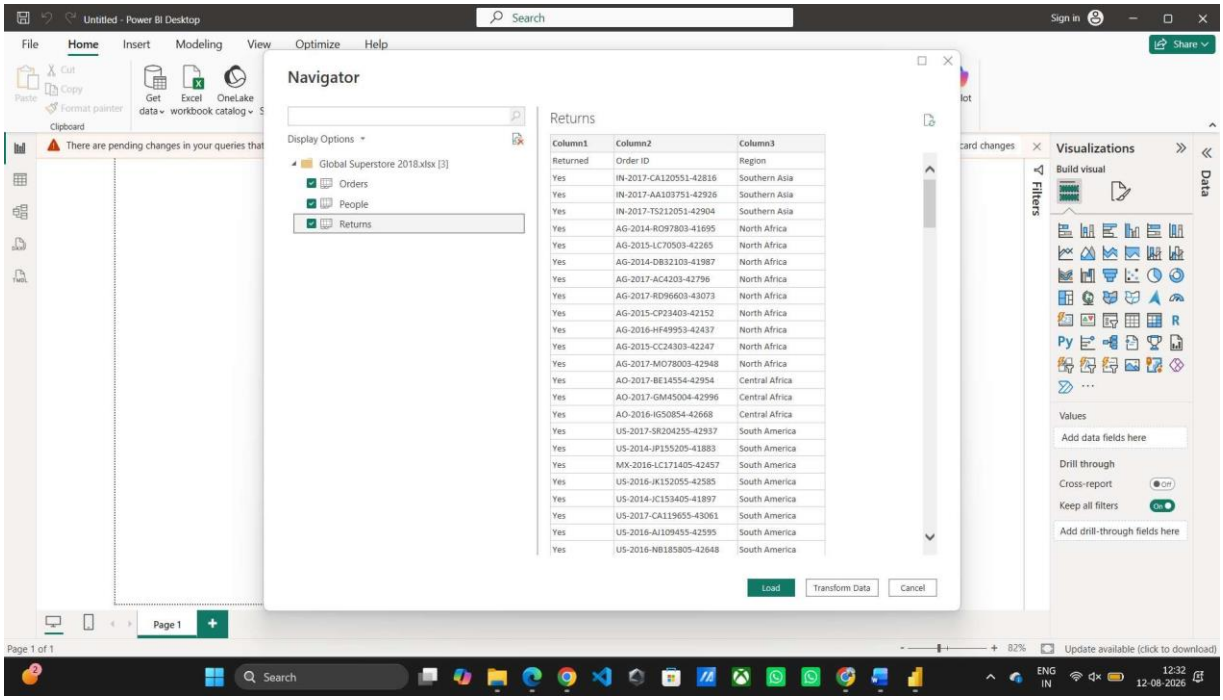
Part A: Dashboard Planning

Step 1: Open the Power BI project created in previous experiments.



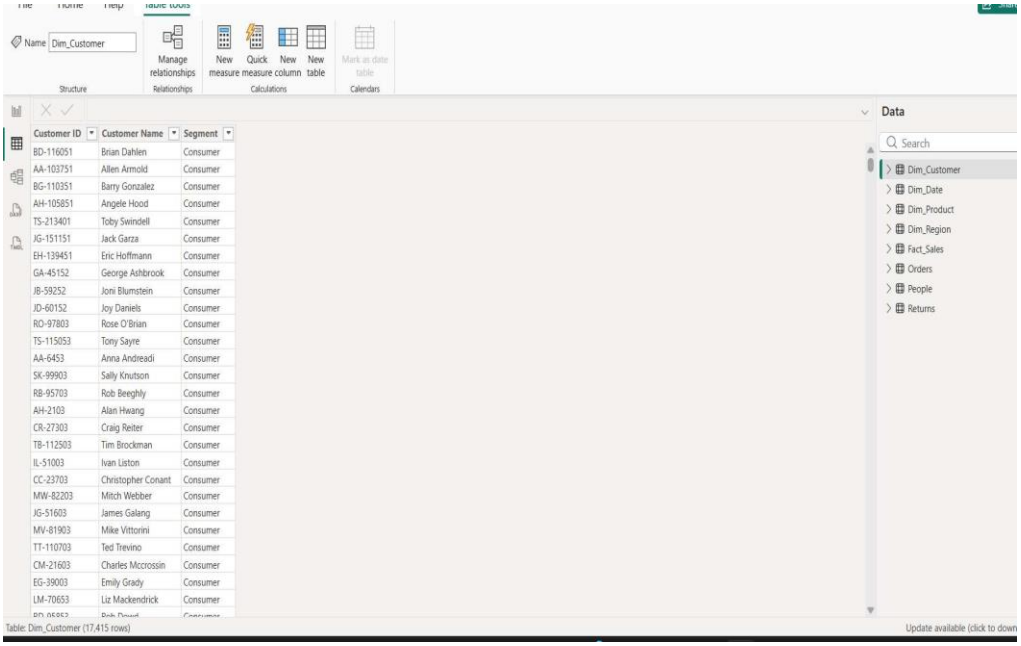
Step 2: Verify the imported tables.

Check the Data/Fields pane and verify that the existing tables, such as `Orders`, `Fact_Sales`, `Dim_Product`, `Dim_Region`, and `Dim_Date`, are available.



Step 3: Create a Table Visual.

1. Open Report View.
2. Select an empty area of the report canvas.
3. Choose the Table visual from the Visualizations pane.



Step 4: Add the required fields.

Add the following fields to the Table visual:

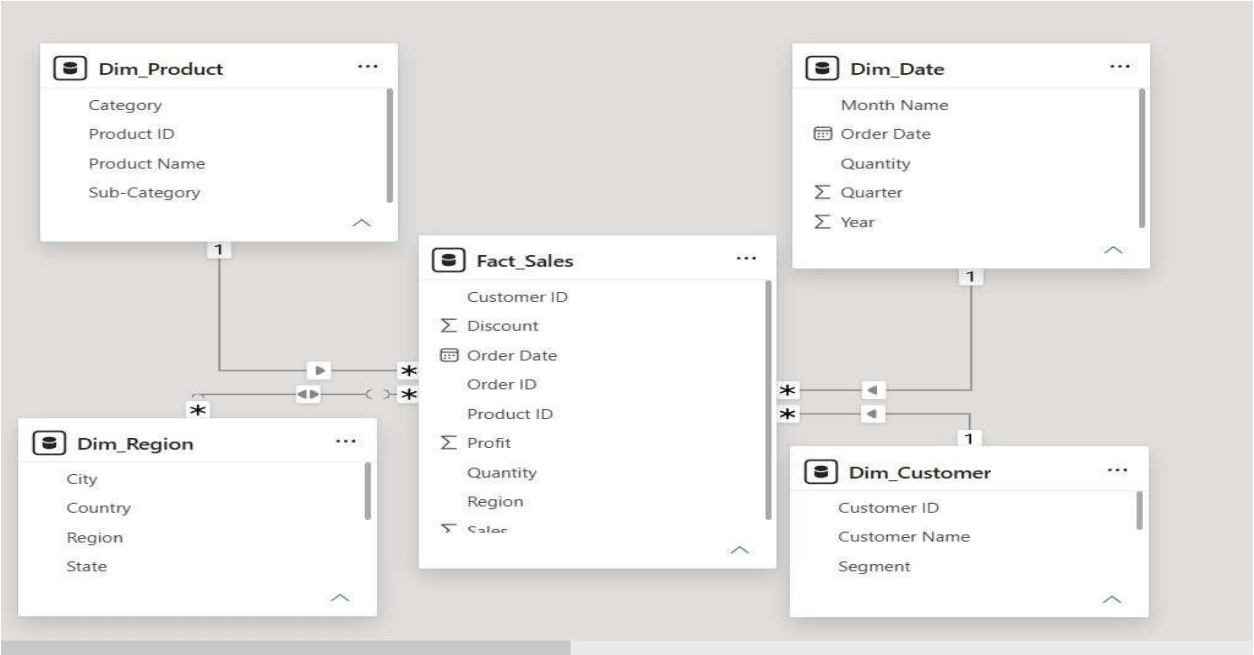
- Order ID
- Customer Name
- Product Name
- Region
- Sales
- Profit

Select the appropriate fields from the existing tables in the model.

Step 5: Identify key business metrics required by management.

Step 6: Select suitable visualizations for presenting performance indicators.

Step 7: Sketch a dashboard layout before implementation.



Part B: Scheduled Reporting Plan

Step 8: Open the Power BI Reports.

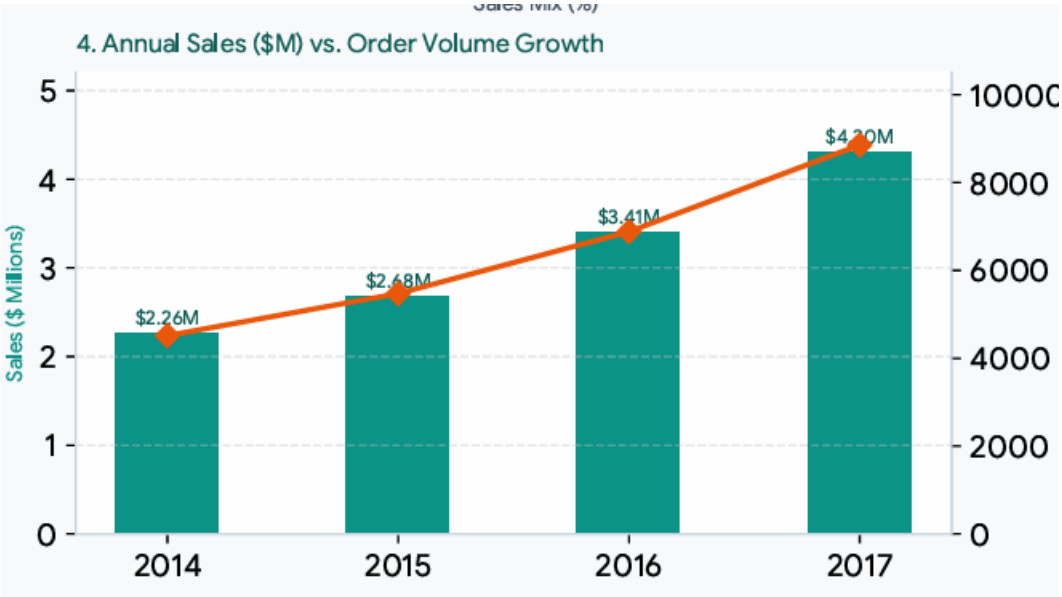
Open Microsoft Power BI Desktop and load the completed Experiment 8 report containing the following pages:

- KPI Monitoring Dashboard
- Monitoring Scenario Analysis
- Daily Sales Report
- Weekly Profit Report
- Monthly Executive Dashboard

Step 9: Identify Reports for Scheduling.

Select the following reports for scheduled delivery:

- Daily Sales Report
- Weekly Profit Report
- Monthly Executive Dashboard



Step 10: Define Reporting Frequency.

Specify the reporting frequency for each report:

- Daily Sales Report – Daily
- Weekly Profit Report – Weekly
- Monthly Executive Dashboard – Monthly

Step 11: Identify Intended Users.

Assign the reports to their respective users:

- Daily Sales Report – Sales Team
- Weekly Profit Report – Finance Team
- Monthly Executive Dashboard – Senior Management

Prepare a Scheduled Reporting Plan based on the report frequency and intended users.

Part C: Automated Distribution Design

Step 12: Prepare a Report Distribution Matrix.

Create a distribution matrix containing the report name, user group, frequency, and delivery method.

Step 13: Identify Delivery Methods.

Select suitable delivery methods for each report:

- Email
- PDF

- Shared Workspace
- Power BI Service

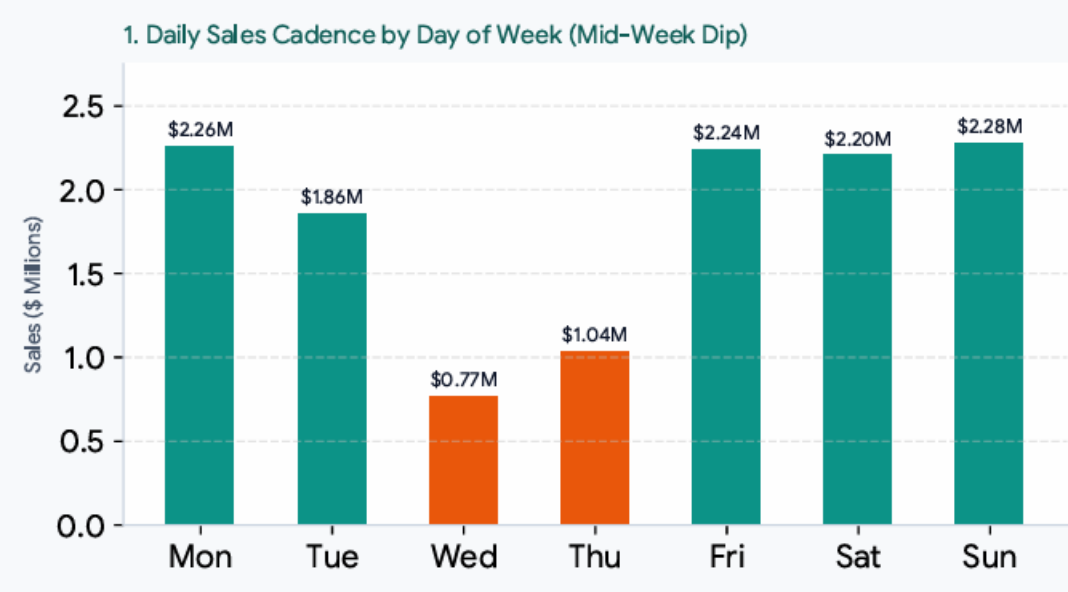
Step 14: Assign Distribution Requirements.

Specify the appropriate delivery method for each report:

- Daily Sales Report – Email
- Weekly Profit Report – Email / PDF
- Monthly Executive Dashboard – Power BI Service / Shared Workspace

Step 15: Finalize the Distribution Matrix.

Organize all report distribution details into a Report Distribution Matrix.



Part D: KPI Alert Configuration

Step16: Identify Key Performance Indicators.

Use the KPI Monitoring Dashboard to identify the following measures:

- Total Sales
- Total Profit
- Total Orders
- Profit Margin

Step 17: Define KPI Alert Conditions.

Specify the following alert conditions:

- Profit Margin below 10%.
- Sales declined greater than 15%.
- Returns increase greater than 20%.

Step 18: Identify Alert Recipients.

Assign suitable recipients for the alerts:

- Profit Margin Alert – Finance Team
- Sales Decline Alert – Sales Manager
- Returns Increase Alert – Operations Team

Step 19: Specify Alert Actions.

Define suitable actions for each alert, such as reviewing business performance, investigating the cause of decline, and taking corrective measures.

Prepare a KPI Alert Configuration Sheet containing the conditions, recipients, and actions.



Part E: Monitoring Scenario Analysis

Step 20: Open the Monitoring Scenario Analysis Page.

Open the Monitoring Scenario Analysis page in Power BI.

Step 21: Analyze Regional Profitability.

Examine the Profitability by Region chart and identify regions experiencing low or declining profitability.

Step 22: Analyze Monthly Profit Trends.

Examine the Monthly Profit Trend chart and identify changes in regional business performance using the Region slicer.

Step 23: Review Regional Performance Summary.

Analyze the Total Sales, Total Profit, and Profit Margin values in the regional performance table.

Step 24: Identify Alert Triggers.

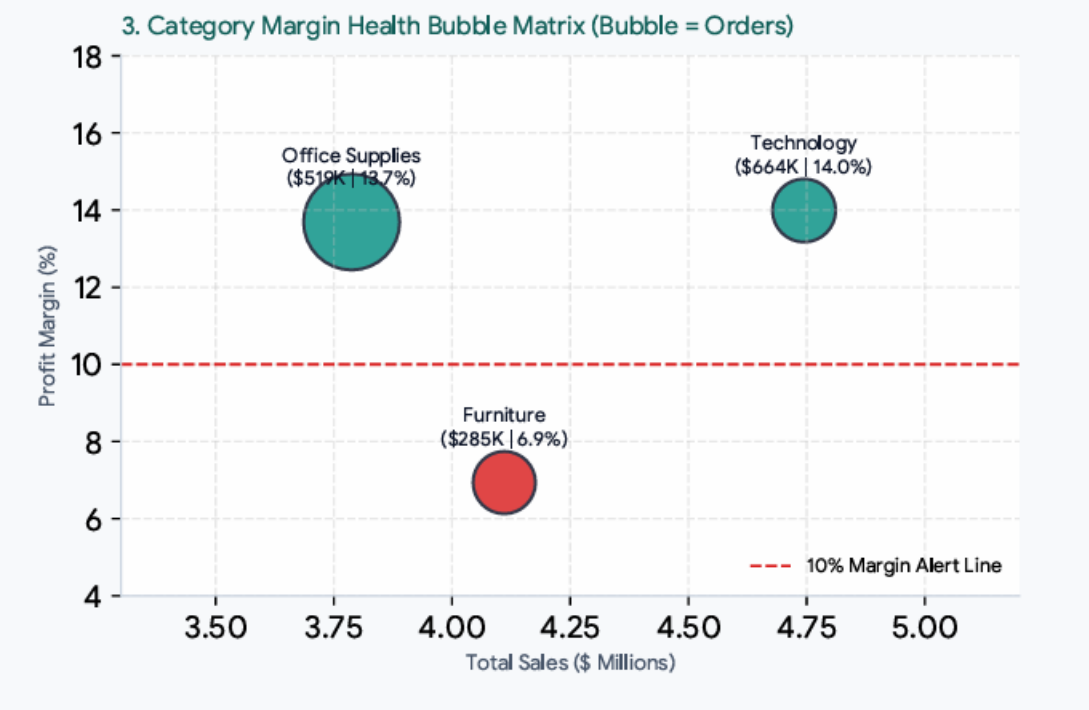
Identify possible alert conditions when a regional branch experiences declining profitability.

Step 25: Identify Corrective Actions.

Suggest corrective actions such as reviewing expenses, improving sales strategies, and focusing on profitable product categories.

Step 26: Record Observations and Capture Screenshots.

Document the monitoring findings, alert triggers, recipients, and recommended corrective actions. Capture screenshots of the Power BI dashboard as supporting evidence.


3. Category Margin Health Bubble Matrix (Bubble = Orders)
Profit Margin (%)
18
16
14
12
10
8
6
4
Office Supplies
($519K | 13.7%)
Technology
($664K | 14.0%)
Furniture
($285K | 6.9%)
10% Margin Alert Line
3.50
3.75
4.00
4.25
4.50
4.75
5.00
Total Sales ($ Millions)

Result

Report scheduling, automation, and alert strategies were successfully designed. Automated reporting plans and monitoring mechanisms were developed to support business operations.

LEARNING OUTCOMES:

- Apply report scheduling concepts in Business Intelligence systems.
- Design automated report distribution strategies.
- Analyze KPI monitoring requirements.
- Evaluate alert mechanisms for business decision support.
- Develop recommendations for BI operations.